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AILIS 1.0: A new framework to measure AI literacy in library and information science (LIS)

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A B S T R A C T

As artificial intelligence (AI) becomes more embedded in academic and professional settings, assessing and improving AI literacy among current and future information professionals is increasingly important. However, research in this area within Library and Information Science (LIS) remains exploratory, and more evidence is needed to guide training and curriculum design. This study assesses AI literacy among LIS students and librarians, highlighting key areas and groups for targeted training.

To this end, the AILIS 1.0 questionnaire was developed from existing AI literacy tools in higher education and adapted to the LIS context with expert input. It was administered to 163 respondents at the Complutense University of Madrid (Spain). Descriptive statistics and non-parametric tests were used to examine gender and group differences. To further validate the findings, three focus groups with LIS undergraduates were conducted.

Functioning, Ethics, and Evaluation emerged as core dimensions of AI literacy. Functioning scores correlated strongly with all other dimensions except self-assessed Usage. Overall, library professionals outperformed students, particularly in Ethics and Usage. However, students, especially first-years, reported higher self-efficacy despite lower performance, indicating a tendency to overestimate their AI literacy, as confirmed by focus groups.

The research underscores the need for educational strategies in AI literacy and greater involvement of educators and professionals. The higher AI literacy shown by librarians should encourage professionals to take more active roles in AI literacy training. Finally, results highlight the potential of AILIS 1.0 as a diagnostic tool, but also as a framework to evaluate AI literacy within LIS.

Introduction

The growing presence of Artificial Intelligence (AI) in higher education poses significant challenges, especially in a sector that European legislation considers to be "high risk" due to the impact that educational decisions may have on people's lives (European Union, 2023: 16). The applications of AI in this area are broad and promising, with special relevance in supporting research and innovation in teaching and assessment methodologies (Gouveia et al., 2023; Xia et al., 2024). However, their adoption also entails notable risks, such as plagiarism, the emergence of new dishonest practices, anxiety, the introduction of biases, inequalities, misinformation, and the decrease in human interactions (Dempere et al., 2023; Walczak & Cellary, 2023). Mental exhaustion, hindered creativity, and over-reliance have also been reported as possible consequences of the introduction of AI in teaching activities (Kumar et al., 2024). Collectively, these challenges contribute to the sense that higher education is "at a tipping point of relevancy and survival" (Bennett & Abusalem, 2024, p. 206). Against this backdrop, universities need to strengthen teachers' and students' intellectual, social, and communicative skills necessary for a responsible and ethical use of AI technologies (Song, 2024).

Currently, research on the appropriate uses of AI in higher education is still ongoing. In the field of Library and Information Science (LIS), the available educational guides prioritize its capacity for data analysis and processing (Torres-Salinas & Arroyo-Machado, 2024), bibliometrics, scientometrics, and webometrics research (Saeidnia et al., 2024), as well as its role in scientific publication and review (Kousha & Thelwall, 2024; Lund et al., 2023). However, the integration of AI in the classroom requires students not only to be knowledgeable of some practical applications, but also to understand the ethical implications of its use, promoting equitable learning environments (Grote-Garcia & Ortlieb, 2024). It is also of fundamental importance that LIS students understand the potential disruptive impact of AI in professional settings (Oladokun et al., 2024).

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AI in libraries

In academic libraries, AI is increasingly used for tasks like cataloguing, information literacy, marketing, and infrastructure management, enhancing user experience and internal operations (Molaudzi & Ngulube, 2025). According to Mupaikwa (2025), AI innovations in libraries have enhanced services by enabling automation, personalization, and greater efficiency. In some cases, robots have been deployed to provide intelligent security and consultation services. However, adoption remains limited due to lack of strategic planning, insufficient training, algorithmic illiteracy among librarians, and ethical concerns such as privacy (Molaudzi & Ngulube, 2025). For some library services, AI chatbots like ChatGPT should still be considered a complement supporting human decisions, because of risks of misuse and the unreliability of responses, among the others (Adetayo, 2023). Additionally, the adoption of AI raises concerns around equality, diversity, and inclusion in a profession facing issues like the gender pay gap, with the potential to reinforce inequalities as men dominate technical and senior roles that are less vulnerable to automation (Cox & Mazumdar, 2024). Even if AI could also support female librarians' professional role and career progression, by improving their personal skills, as in Edam-Agbor et al.'s (2025) research on academic libraries in Nigeria, it is still early to fully understand AI's impact on LIS professions. In this sense, not only should AI technologies in libraries be considered still in their infancy, but their impact is still unpredictable as their introduction depends on technical developments that are beyond the control of librarians (Cox, 2023; Cox & Mazumdar, 2024).

Librarians adopt a wide range of positions toward technology, with attitudes spanning from avoiders or "anti-technology traditionalists" to experts capable of developing technological solutions (Cox, 2023). Apparently, most librarians do not participate in AI training, are not involved in AI-related decisions, feel uncertain about their job security, and have concerns about the reliability of AI and the misuse of user-generated data (Mupaikwa, 2025). While librarians show potential for upskilling, the full benefits of AI can only be realized through well-structured training programs that afford them the skills needed to innovate and integrate AI in library routines (Lo, 2024; Molaudzi & Ngulube, 2025). Notably, in this emerging technological era, AI literacy courses may become one of the first AI-related services provided by librarians (Cox & Mazumdar, 2024). This view is also reflected in the IFLA Statement on Libraries and Artificial Intelligence (2020) and Padilla's (2019) framework of critical domains that libraries should address when engaging with AI, machine learning, and data science. Indeed, some libraries are training in AI-related technologies and some professional associations have provided guidance and guidelines for AI use (Ali & Richardson, 2025; Bongiovanni et al., 2024). The real scope of these interventions and their actual presence, however, are unknown. While librarians are expected to provide guidance and training in using AI technology, the truth is that professionals need to receive training themselves (Ali & Richardson, 2025). This need for structured training is supported by the OECD Recommendation on Artificial Intelligence (2025), which (although not specific to libraries) offers a relevant framework for LIS. Librarians providing instructions surveyed in Bongiovanni et al. (2024) reported learning about AI mostly through self-directed learning, with secondary strategies being professional development opportunities, formal education, research and scholarly activities, and peer collaboration. Limited school training in AI could explain at least partially poor adoption of AI related technologies in libraries, while research in AI literacy is surprisingly scant especially regarding librarians' instruction (Bongiovanni et al., 2024). The limited training opportunities specific to the field may also explain AI literacy being highly variable in the LIS discipline (Hofman et al., 2024).

Purpose of the research

This research presents a questionnaire developed specifically for LIS higher education with the purpose of measuring student's literacy of AI and guiding in identifying areas in which they may be more in need of further instruction. Pinski and Benlian (2024) criticize the limited research on AI literacy among expert adult users, including those in the LIS field, noting that most studies focus on K-12 students, adults in personal contexts, or families. Additionally, previous research into AI literacy in LIS higher education has been mainly exploratory (Hossain et al., 2025; LaFlamme, 2025), theoretical (Shiri, 2024), or has conceived AI literacy simply as an improvement of information literacy by strengthening human-AI collaboration via prompting engineering (Lo, 2023). Existing studies either gather empirical data or explore theoretical issues. Lo's (2024) survey gathers mixed data from academic librarians across various dimensions including self-assessment of AI understanding, AI-related processes, such as participation in AI-related discussions, and opinions on the application of AI in library contexts. However, this study does not address specifically AI literacy. Shiri (2024) classifies frequently used words and phrases in the literature about AI literacy into a taxonomy, with no support of further empirical user data. In this sense, AI-literacy specific problems and challenges in LIS have not been paid due attention.

The questionnaire, named AILIS 1.0 (*Artificial Intelligence Literacy in Library and Information Science*), was developed based on previous instruments specifically intended for application in higher education and was adapted to the LIS area based on previous relevant literature and expert judgement. The competencies that it covers are also relevant for the profession both because today's undergraduates will be future librarians and because academic librarians are expected to be AI literate and provide AI instruction themselves. AILIS 1.0 was administered to a sample of 100 students of different LIS programs of the Complutense University of Madrid and 63 academic librarians of the same institution. Results are intended to inform on the central topics of AI education and identify gaps in knowledge and competencies, both in students and librarians. Additionally, in the case of librarians the questionnaire aims to provide evidence about their preparedness as instructors and trainers in AI. Participants are expected to perform better in softer competencies, such as those related to ethics, with a longer trajectory in LIS, than in harder, more technical competencies, such as those related to functioning, as reported in previous research (Hornberger et al., 2023; Hossain et al., 2025; Lee et al., 2024; Toker Gokce et al., 2024). Similarly, performance is expected to reflect the development of participants' cognitive dimensions as they progress academically, as observed in Katsantonis and Katsantonis (2024), where students at more advanced stages demonstrated a deeper understanding of the significance and added value of AI. Additionally, a gender analysis will also be performed considering the high prevalence of females among LIS students and librarians, and the auto-efficacy gender gap typical of technology use (Cai et al., 2017). In Ofosu-Ampong (2023), gender influenced the use of AI-based tools, with male students using them more frequently than females, and showing greater familiarity and more positive attitudes toward it. Sentiments toward these new technologies are also shaped by gender, and males perceive AI technologies to boost their creative thinking, whereas females value the potential of AI to achieve higher competencies in their studies (Al-Samarraie et al., 2024). Although AI technologies are expected to impact women's employment opportunities and workplace experiences, more research is needed to fully understand these effects (Collett et al., 2022). Finally, the results obtained from the administration of the questionnaire together with qualitative feedback obtained from three focus groups are intended to provide data about its improvement and further refinement.

Literature review: AI literacy

According to Long & Magerko (2020: 2), AI literacy gathers *"a set of competencies that enables individuals to critically evaluate AI technologies; communicate and collaborate effectively with AI; and use AI as a tool online, at home, and in the workplace"*. Several authors agree that AI literacy is

closely linked to other forms of literacy, such as information literacy (Akakpo, 2024; Lo, 2023), digital literacy (Biagini et al., 2024; Wang et al., 2023; Wolters et al., 2024), and data literacy (Chiu et al., 2024; Long & Magerko, 2020). Cox and Mazumdar (2024) argue that data and AI literacy as a combined concept should be considered a dimension of information literacy and broader "citizen literacies," given the widespread and accessible use of AI-based technologies from search engines and recommendation systems to translation applications and grammar or style checkers. Sometimes, AI literacy is understood as part of digital literacy (Lee & Park, 2024), whereas at other times digital literacy and AI literacy are considered to intersect with one another (Wolters et al., 2024). Under the European DigComp Framework 2.2, AI is conceived as any other technology, though characterized as "new and emerging" (Vuorikari et al., 2022). In library contexts, the understanding is that Information literacy enables a critical, thoughtful, and ethical interaction with AI systems, being for this reason its fundamental constituent (LaFlamme, 2025; Lo, 2024). Ridley and Pawlick-Potts (2021) understand that integrating "algorithmic literacy" – i.e., the ability to understand, interpret, create, and critically evaluate algorithms and their societal impacts - into information literacy can afford the awareness, understanding, and skills needed for making meaningful use of AI and enable "explainable" AI. Chiu et al. (2024), on the other hand, highlight that AI literacy overlaps with data literacy, although the boundaries between the two are not fully defined, requiring further research. Long and Magerko (2020) link AI literacy competencies with those established by Calzada Prado and Marzal (2013) for data literacy, highlighting the key role of data in understanding machine learning. According to Olari (2023), the intersection between both literacies lies in the ability to critically interpret big data and in the development of statistical literacy skills. In educational contexts, AI literacy and data literacy are often considered intertwined. Picasso et al. (2024), for example, conceive of both as essential assets in experiential learning activities and cultivating critical thinking skills. Data literacy as a core component of AI literacy is stressed by Cox (2021) together with computational thinking, soft skills and "fusion skills", i.e. all those skills that allow humans to work meaningfully with computers. In the framework of academic libraries, AI affordances can shape new users' experiences, by enhancing information discovery and personalization, and boosting research productivity, therefore capitalizing on users' digital literacy and information evaluation research capabilities (Meakin, 2024).

Indeed, AI literacy is understood as a complex, multidimensional construct that requires learners to be knowledgeable of both the human and technological dimensions of AI in order to make a "safe, ethical and meaningful use of AI in education", as the UNESCO's guide explains (Holmes & Miao, 2023: 25). This multidimensionality has been articulated by Shiri (2024) in a faceted taxonomy with 13 dimensions encompassing conceptual, technical, ethical, social, and applied aspects of AI literacy. For its multiple connections with other forms of literacy, Černý (2024) conceives of AI literacy as a fundamental multidimensional competency both in everyday life and in the labor market. In this line, Huang et al. (2023) consider it a subset of critical thinking, understood as a broader transversal competency. Despite the various theoretical approaches, there is still no clear consensus on its definition, and research in higher education has focused mainly on evaluating the impact of AI training programs, with specific AI literacy scales more common for the general population than higher education students (Hornberger et al., 2023; Laupichler, Aster, Perschewski, & Schleiss, 2023; Lintner, 2024). As Laupichler et al. (2022) point out, the study of AI literacy is still in its early stages, so it is crucial to define its foundations within higher education and establish the essential contents for its teaching. Most instruments have been developed with a general focus in mind, with no adaptation to specific fields or knowledge areas, though there are some exceptions, especially in the health sciences (Iwasawa et al., 2023; Karaca et al., 2021; Laupichler et al., 2024; Wolters et al., 2024). According to Pinski and Benlian (2024), the multidimensional, holistic and complex nature of AI literacy urges identifying specific facets unique of AI literacy, beyond its obvious overlap with digital and media literacy.

Methodology

The research includes data collected via the administration of AILIS 1.0 to LIS students and librarians of the Complutense University of Madrid (Spain), and three focus groups with some of the students who took the test. The focus groups were carried out after the administration of AILIS 1.0, and allowed to further validate the questionnaire and direct future actualizations of the instrument.

Development of the instrument

Validity

According to the *Standards for Educational and Psychological Testing*, the validity of an instrument measures the extent to which theory and evidence support the interpretation of the test scores (American Educational Research Association (AERA) et al., 2014). In this regard, the validity of AILIS 1.0 is supported by multiple sources of evidence, including prior research on AI literacy and related tools, expert judgement from the LIS field, and student feedback gathered during both the questionnaire's development and the validation of its results.

Identification of competencies for AI literacy. The questionnaire was developed based on a review of the literature and previous instruments used with undergraduates. The exploratory review of the literature was carried out in Scopus in April 2024, using the following search strategy: ("artificial intelligence" AND (literacy OR competencies) AND ("higher education" OR "university students")), and limiting results to the 2022–2024 period, that is, since the availability of the generative AI application ChatGPT to the general public. The results were limited to studies that explicitly addressed AI literacy, presented some type of empirical research with undergraduates, or dealt with specific competencies, such as ethics or emotional intelligence. Additionally, searching the bibliographic references of the 8 papers that were finally selected, 9 additional papers were added to the initial corpus – spanning the period 2020–2022. Studies focusing on tools of lesser relevance in the higher education context, such as Siri and Alexa, for example, or on aspects not directly related to AI literacy, were excluded. For example, Sindermann et al. (2021) and Schepman and Rodway (2020) measure "attitudes" rather than knowledge and literacy. On the other hand, a contribution that addressed the problem in secondary school students was included (Ng et al., 2023), as it developed previous work with great influence on subsequent research in academic settings (Ng et al., 2021). Similarly, Carolus et al. (2023) was included in the analysis, for addressing psychological competencies especially relevant in professional contexts. A total of 17 studies were selected, 9 of which developed specific questionnaires designed to measure AI literacy. All are listed in Annex I and included in the Reference section. Among the reviewed scales, two (Hornberger et al., 2023; Lee & Park, 2024) were considered to have the most robust evidence in the context of higher education, according to a systematic review of AI literacy scales (Lintner, 2024).

In AI literacy scales, the elements that make a person competent are usually classified into large dimensions, often, though not always, based on results from factorial analysis (Carolus et al., 2023). Such dimensions may be called "themes" (Long & Magerko), "factors" (Lee & Park, 2024), or can even be conceived as independent "constructs" or "sub-constructs" (Wang et al., 2023). The sources consulted may use different names for apparently equal dimensions. For example, the dimension that in AILIS is called "Use of AI" appears as "Operational dimension" in Biagini et al. (2024), "Use & apply AI" in Ng et al. (2021), "Apply AI" in Carolus et al. (2023), and "Ai Usage " and "Human actors in AI" in Pinski and Benlian (2023). Different instruments may classify competencies

into dimensions using criteria that are inconsistent with those of other sources. In this research the selection of dimensions and corresponding competencies was based on expert human judgement, supported by the mentioned literature review. Firstly, 5 dimensions were identified by 7 LIS experts (6 lecturers, including the authors, and the director of a library specialized in LIS) in a specific meeting: What is AI, Functioning, Usage, Ethics, and Evaluation. The 71 competencies collected under these dimensions from the 17 selected works were evaluated by the experts via two online forms (May 2024). Following several meetings to agree on a common purpose, the online forms afforded individual assessment of the 71 selected competencies according to their importance for LIS training on a scale from 1 (not at all important) to 5 (very important), as well as the provision of specific observations. In selecting potential items from existing scales and previous literature on relevant competencies, as well as in gathering experts' evaluation via a digital form, we proceeded similarly to Chang and Kuo (2025) who develop a scale for measuring digital literacy. As a result of these consultations, a set of 38 competencies was selected based on each item's Content Validity Index (CVI) and Coefficient of Variation (CV). Following the methodology of Lee and Park (2024), only competencies with a positive CVI and a CV of 50 % or less were included. The development of the questionnaire is thoroughly described in previous work (Authors, submitted).

Questionnaire development, disciplinary adaptation and students' feedback. After the identification of core competencies, specific questions were developed for each of the major dimensions. This process was carried out through a series of face-to-face meetings and interactions around content shared on the cloud, relying also on selected disciplinary literature, in order to guarantee the alignment between LIS students' learning standards and test content (American Educational Research Association (AERA) et al., 2014). Though the meetings were not recorded, the comments appended to the drafts were recorded and the decisions taken were applied to obtain new shared drafts of the questionnaire. Disciplinary literature on AI was used to establish the type of knowledge and competencies expected in LIS, including Frické (2024), IFLA's suggested resources (2023), and contributions to the topic such as Cox (2021) and Cox and Mazumdar (2024). Specific research on libraries' main concerns regarding AI use were also included, in particular on ethical issues, learning analytics and user-generated data (Cox, 2022; Jones & Hinchliffe, 2023; Padilla, 2019), which resulted in 22,90 % data-related competencies in the final instrument. The high proportion of data-related competencies in AILIS 1.0 differentiates it from the nine previous instruments or studies, which allocate between 0 % (Carolus et al., 2023) and a maximum of 17.6 % (Long & Magerko, 2020) to such competencies, as reported in the authors' previous work (Authors, submitted). AILIS 1.0 also differs from Lo's (2024) framework for AI literacy in academic libraries, which focuses solely on data-related issues concerning privacy and security. The potential applications of robotics in libraries, as described by Cox (2021), were excluded, as they were considered less relevant than broader competencies in the context of higher education. Similarly, specific competencies of "media literacy" were not explicitly included, despite IFLA FAIFE (2020) considering media literacy a specific part of AI literacy, because the focus of the instrument was on AI competencies and training needs in this specific area.

While defining the specific questions that would be used to measure each competency, the competencies themselves were redefined in a process of progressive filtering, adaptation, and disciplinary alignment. As a result, some competencies were nuanced. For example, the competency "knowing how AI perceives the data entered" was split into two different items: "knowing how AI perceives the data entered for model training", on the one hand, and "knowing how AI perceives the data entered by users", on the other. Also, new competencies were added specifying those defined in previous models, for example, "understanding bias in data", or reflecting concerns about the environmental impact of AI and the impact of AI on people with functional diversity, in line with UNESCO's recommendations about inclusion, equity, and diversity (Holmes & Miao, 2023). No specific competencies were introduced regarding gender, though the gender perspective was taken into account in the formulation of specific questions. Two questions were adapted from a previous instrument (Hornberger et al., 2023).

The process of disciplinary adaptation yielded a preliminary set of 39 competencies and corresponding questions, which were further evaluated by 7 students via an online form. Students were asked to take the test, and at the same time provide suggestions for modifications, express doubts, and rate each item relevance for AI literacy on a scale from 1 to 10 - a format that they preferred due to its similarity to the scoring systems they are accustomed to. Global students' feedback, including their comments, score, and rating was used to filter out 4 competencies, which resulted in a final questionnaire of 35 questions in 5 dimensions (Table 1), each related to a specific competency. The entire set of questions and the respective competencies is available as Annex II at the end of the article.

Scoring system

The scoring system combines multiple-choice questions with a single correct answer and three self-assessment items rated on a scale from 1 to 5. All self-assessment questions are included in the Usage dimension, based on the understanding that multiple valid uses of AI may exist. This mixed scoring approach was adopted following a careful review of the relevant literature and drawing on the authors' classroom experience. Self-assessment, as the standard measurement in most of the reviewed questionnaires, assumes that perceived behavioral control and self-efficacy are important predictors of voluntary behavior and actual performance (Carolus et al., 2023; Hornberger et al., 2023). Additionally, attitudes toward technology can shape individuals' actual use of it (Wang et al., 2023). The *Technology Acceptance Model* (TAM) makes technology use and acceptance dependent on perceived usefulness and perceived ease of use (Chan & Zhou, 2023). However, sometimes it is actual knowledge that changes students' attitudes and perceptions (Korte et al., 2024). An AI course centered around vocabulary, workings, and ethics, among the others, proved successful in changing students' perception of AI, both by fostering a more critical stance toward GenAI and by removing concerns over its unethical use (Ngo & Hastie, 2025). Similarly, in Tzirides et al., 2024 students reported feeling more comfortable using GenAI tools after a 3-weeks course experimenting with actual AI applications. AI explainability, i.e., being able to explain how AI algorithms work and how they generate responses, may influence different users' attitudes toward AI systems, such as trust or acceptance (Wen et al., 2024). In addition, self-perception or self-efficacy and actual knowledge may not always be aligned. Chang and Kuo (2025) found a misalignment between scores on "skill" dimensions and scores on "attitude" dimensions after applying a digital literacy scale to primary and secondary school students. Perceived AI literacy may also vary according to demographic factors, especially gender, which is of special importance in LIS, an area with high prevalence of females among students and professionals. If in general females' attitude toward technology use appears to be less favorable compared to males,

Table 1
Items by competency of the instrument.

Abbreviation	Dimension	No. of items
Dim1	What is AI	4
Dim2	How AI Works	13
Dim3	Usage of AI	5
Dim3a	Usage – self- assessment (scales)	3
Dim4	Ethics of AI	7
Dim5	Evaluation of AI	3
	Total items	35

as emerged from a meta-analysis covering the period 1997 to 2014 (Cai et al., 2017), a gender gap has been detected specifically with AI. The administration of the ATAI (*Attitude Toward Artificial Intelligence*) scale, measuring acceptance and fear of technology, returned greater interest and more positive attitude toward technology in male participants (Sindermann et al., 2021). Similarly, the application of PAILQ-6 scale returned a similar picture, as scores were found to be significantly lower in female participants compared to male counterparts (Grassini, 2024). Besides, the heavy reliance on self-assessment makes especially urgent to develop alternative assessment tools able to provide stronger evidence for training programs (Wolters et al., 2024).

AILIS 1.0 scoring system comprises both dimension-specific normalized scores (percentages) and a global weighted normalized score. For each participant, the number of correct answers in each dimension was converted into a percentage, creating dimension-specific normalized variables. The global score was calculated by weighing each dimension according to the number of items it contained (see Table 1), in order to ensure comparability across the different dimensions of the AILIS 1.0 instrument. The three self-assessment items related to AI Usage were treated as a distinct dimension and normalized to the same 0–100 scale as the other dimensions, using the formula $(X - 1)/4 \times 100$, where 1 maps to 0 and 5 maps to 100. This allowed meaningful comparison and integration into the global score.

The final Weighted Normalized Score (WNS) was computed by applying the following formula, in which each dimension (Dim1 to Dim5, including Dim3a) represents a normalized value, rescaled to range from 0 to 100:

$$WNS = \text{Dim1}^{*}\ 0,11 + \text{Dim2}^{*}\ 0,37 + \text{Dim3}^{*}\ 0,14 + \text{Dim3a}^{*}\ 0,09 + \text{Dim4}^{*}\ 0,20 + \text{Dim5}^{*}\ 0,09.$$

Based on the WNS, participants were categorized into three AI literacy levels: Low performance (WNS < 50), Medium performance (WNS ≥ 50 and < 70), and High performance (WNS ≥ 70), in order to facilitate comparative analyses across groups.

Reliability and error measurement

To assess the reliability of the questionnaire and ensure that it returns consistent results, two administration procedures were conducted with a group of 17 students, one week apart, paralleling the administration of the questionnaire to other participants. The 17 participants did not receive any formal training during the week between the test and the retest, they were instructed to refrain from searching for information related to the questionnaire, and were informed that the training was scheduled to take place after the retest procedure. On both occasions, the questionnaire was administered during class hours and in the same format. Unweighted Cohen's Kappa for test-retest reliability was applied to individual dichotomous responses (correct / incorrect), while weighted Kappa with equal weights was computed for ordinal data corresponding to the self-assessment scale questions in the dimension of Usage (Sainani, 2017). Additionally, Intraclass Correlation Coefficient (ICC) was computed for the global score (*WNS)* and individual scores obtained in each dimension. All reliability measures were computed using the *R package Irr* (Gamer et al., 2019). The settings for the ICC included a two-way model (considering that the responses were provided by the same students), and the type of agreement measured was "agreement", looking at the similarity of values more than mere consistency among scores. Additionally, error measurement was quantified as Standard Error Measurement consistency (SEM consistency), calculated using the standard deviation of the differences between the two measurements (SDdifference) and adjusting for the two measurements with the formula SEM consistency = SD difference / $\sqrt{2}$. These reliability statistics were selected among those recommended by the COSMIN guidelines (Mokkink et al., 2019) and additional sources (De Vet et al., 2011; Leppink & Pérez-Fuster, 2017), and they were preferred over more common measures of reliability often used for AI scales, especially Cronbach's alpha (Lintner, 2024).

Questionnaire administration and analysis

The questionnaire was administered between November 2024 and beginning of March 2025. Students' responses were collected during class hours, whereas academic librarians were invited via email at the beginning of February 2025. In both cases, it was completed online and allowed participants to revise their scores and obtain a copy of their responses, which motivated to complete the questionnaire and generated results with practically no missing values. In the few cases where missing values were present, responses were excluded from specific analyses where complete data were required, while included in others where appropriate. Before collecting data from participants, ethical approval from the UCM Ethical committee was provided. Demographic data are available in Table 2. The two samples represented 28,6 % approximately of LIS undergraduates and masters' students at the UCM (UCM, 2024), and 18,9 % of UCM active librarians (BUCM, 2024). While most students completed the questionnaire in the classroom at the suggestion of their teachers, librarians chose to do so voluntarily, which may have influenced the scores obtained in favor of librarians.

Before the analysis, some responses were filtered out, two for being repeated (in this case the first set of responses was maintained), and one for suspicion that the questionnaire was not carried out with due attention.

Two researchers independently analyzed the data using descriptive and correlation statistics, then compared their results to ensure consistency. When comparing groups of participants based on gender, prior training, and professional role (students versus librarians), the null hypothesis of no differences was tested using appropriate non-parametric methods, due to the non-normal distribution of data (the Wilcoxon Mann–Whitney test or the Kruskal–Wallis test). For this purpose, Base R and some additional packages have been used, with *corrplot* being particularly relevant (Wei et al., 2024). Dalgaard (2008) was used for guidance on statistical testing in R. AI tools, including ChatGPT and Copilot, were used as supplementary support for coding tasks.

Table 2
Participants' demographics.

Students per year of study	Females	Males	Prefers not to tell	Total
First-year undergraduates	11	7	–	18
Second-year undergraduates	15	6	2	23
Third-year undergraduates	4	3	–	7
Four-year undergraduates	21	9	2	32
Master's students	10	7	1	18
Missing values	2	–	–	2
LIS students - total	63	32	5	100
Librarians	42	19	2	63
Total general	105	51	7	163

Focus groups

After the questionnaire was administered, three focus groups were carried out with 24 of the participating students, providing additional evidence on the validity of AILIS 1.0 and suggesting future improvements. At the time of the focus groups, all participants had used AI to complete various class tasks, including code generation, text processing, reflective questioning, and support in completing classroom quizzes. Focus groups were based on a predefined set of questions organized in 4 major sections: Reflections on questionnaire results, Experience using AI in class tasks, Professional readiness with AI, and AI in university curricula. They were led by two teachers and authors of this paper. The full set of questions is available as Annex III. Two focus groups were audio-recorded and transcribed, whereas in one case participants preferred not to be recorded, and a summary of the main themes discussed including some direct quotations was elaborated immediately after the session (Table 3).

Thematic analysis was performed on the transcripts and summary of the focus groups in order to triangulate data obtained from the questionnaire. The results are a reflexive account shaped by the teaching experience of the coders and not a simple summary of topics (Braun & Clarke, 2023), intended to further validate the instrument, suggest possible improvements of it, and guide in improving AI literacy in the classroom.

Coding of text transcripts was applied by two of the researchers who are also experienced teachers and had witnessed AI usage within the class. The only predefined codes were those corresponding to the different dimensions of the questionnaire (Table 1). The rest was constructed inductively at the intersection between researchers' experience and the text in several rounds of analysis, which is common practice in thematic analysis compared to content analysis (Neuendorf, 2018). Data analysis followed the procedures indicated by Terry et al. (2017), with themes being defined in multiple interactions with the text and meetings among the coders. After a first stage of familiarization with data, text fragments were coded independently on paper. Codes were assigned any time some meaningful concept was identified by the coders, and coded fragments were either entire speaker turns, part of them, or even single words. Practically, all speaker turns, except teachers' words or discourse markers, have been coded with at least one code. Later, the codes identified by the two researchers were transferred to the software for qualitative analysis MAXQDA (24.9.0) in order to look at their reciprocal interactions and to facilitate theme construction. Instead of searching for interrater agreement, all codes were summed up in order to take advantage of the multi-subjectivities and different experiences of the two coders (Braun & Clarke, 2023). As a result, many text segments were assigned multiple codes, a feature that was later utilized to support code visualization (Annex IV) and the construction of overarching themes. During the process, some codes were merged, suppressed, or renamed, until four major themes were identified. Quotations from focus groups transcriptions are attributed to participants by their initials.

Table 3
Summary of main focus groups details.

Group	Date	Participants	Duration	Transcript word count
Group 1	05/12/2024	5	0:22:26	5281
Group 2	09/12/2024	11	0:28:32	4322
Group 3	17/12/2024	8	0:30:00 aprox.	1352*
	Total	24	1:20:58	10,955

* Word count corresponds to a summary of the focus group.

Table 4
Reliability statistics.

Dimensions	ICC	95 %-Confidence intervals	SEM Consistency
WNS	0.499	-0.449 < ICC < 0.821	9.38
What is AI	0.765	0.302 < ICC < 0.917	12.6
Functioning	0.69	0.129 < ICC < 0.889	11.3
Usage	0.0936	-1.78 < ICC < 0.683	23.4
Usage (scales)	0.881	0.677 < ICC < 0.957	8.7
Ethics	−0.352	-3.297 < ICC < 0.533	17.6
Evaluation	0.73	0.267 < ICC < 0.902	19.1

Results

Reliability

The average Cohen's kappa value for the whole instrument was 0.35 with SD = 0.26, and confidence intervals ranging from 0.26 to 0.43, which, according to the recommendation reported in Sainani (2017), can be considered fair to poor reliability (Table 4).

Conversely, the ICC values for each dimension and the *WNS*, as a global measure of reliability, indicate a moderate level of agreement, although the 95 % confidence interval (−0.449 to 0.821) indicate wide differences in consistency among the 35 items. ICC values across the different dimensions vary considerably and are notably low in two softer dimensions - particularly Usage and Ethics. In some cases, the lower bounds of the coefficients are negative, which may be due to the small sample size, but could also reflect participants' fatigue or disengagement from completing a lengthy test twice within an eight-day interval. Questions about Usage and Ethics were proposed in the second half of the instrument, after What is AI and Functioning, which may have found participants especially fatigued. Moreover, inconsistencies in responses within dimensions such as Ethics and Usage may suggest uncertainty or a lack of clarity regarding best practices in these aspects of AI. Meaningfully, the average *WNS* is slightly lower for the test than the retest (average = 57.56 vs 56.46, SD 8.67 vs 13.54), despite a reasonable expectation of improvement. The error measurement for the *WNS* was computed based on SD of the variance between the test and the retest scores, adjusted, as explained in (De Vet et al., 2011: p.111). The resulting SEM consistency is 9.38, which represents the amount of variation in subjects' total scores between the test and retest. In this sense, it reflects a moderate level of error relative to the total score range.

Global results

Fig. 1 shows the distribution of scores for the whole sample across the five different dimensions, as well as for Usage questions measured as scales. Data are more evenly distributed in the Ethics dimension, while the greatest variation appears in the What is AI and Usage dimensions. The highest scores were obtained in Ethics and Evaluation.

According to the correlation matrix of scores across all dimensions (Fig. 2), scores in the Ethics dimension show a significant positive correlation with all other dimensions, except for Usage (scales), where the correlation is significant, but very weak and negative (Spearman rho = −0.17, $p = 0.03$). In a similar way, Functioning correlates positively with all the other dimensions, except Usage (scales).

The strongest correlation corresponds to Functioning with Ethics (Spearman rho = 0.5, $p = 0.00$) and Evaluation (Spearman rho =0.43, p = 0.00). In both cases, correlations can be considered moderate. Meaningfully, scores corresponding to self-assessment (Usage scales) do not correlate significantly with any other dimension, including the Usage dimension, and excluding Ethics. Evaluation scores correlate positively with Ethics (Spearman rho =0.33, p = 0.00), but especially with Functioning.

Overall, AI literacy levels based on the total WNS reveal a predominant presence of moderate proficiency among participants (students

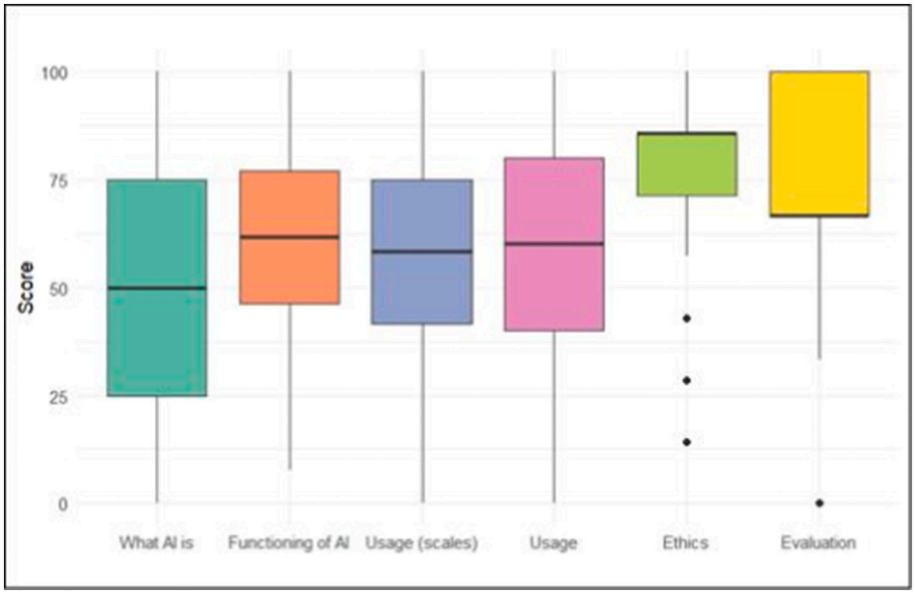

Fig. 1. Weighted normalized scores in all dimensions including Usage (scales).

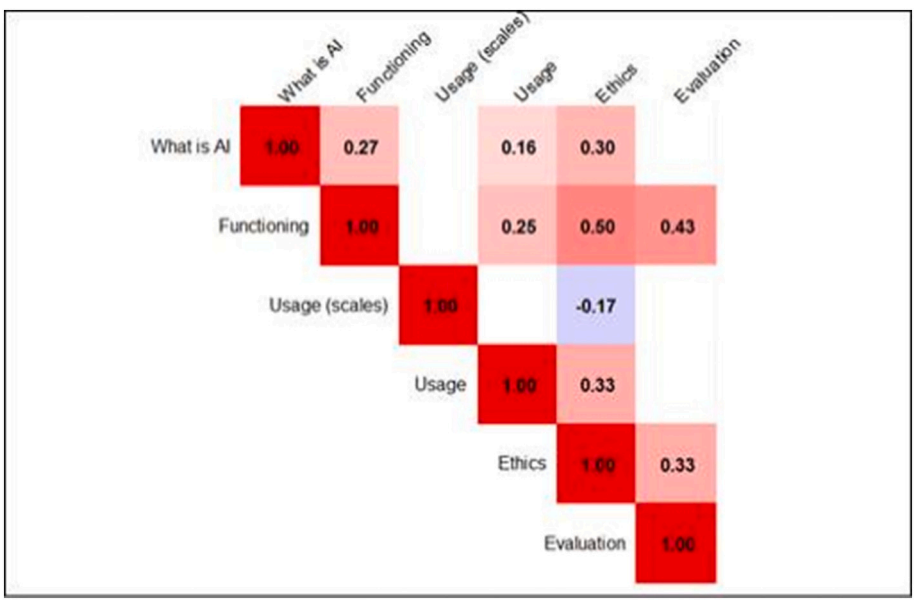

Fig. 2. Correlation matrix of the 5 dimensions plus Usage (scales). Blank cells indicate non- significant correlations ($p > 0.05$). Spearman rho values reported for significant correlations.

and professionals). Specifically, 49.7 % of respondents fall into the medium AI literacy level, while 32.5 % are categorized at the high level. In contrast, 17.8 % of participants demonstrate a low level of AI literacy. These findings suggest that while almost half of the participants show a solid understanding of AI concepts, a significant proportion of the respondents still presents limited knowledge.

Results by groups

Comparing scores in students, advanced students and librarians

In this section, scores are compared between various participants' groups. Students were classified into two categories: first-year to third-year students (referred to as 'students', $n = 48$) and fourth-year and master's students (referred to as 'advanced students', $n = 50$), and compared to librarians. Fig. 3 shows that librarians, and occasionally advanced students, perform better than students in most dimensions, with a few exceptions.

While there are no significant differences in scores obtained in What is AI (Table 5), students, especially first-years students, show a significantly higher self-efficacy when using AI – according to scores obtained in Usage (scales). In Usage and Ethics librarians perform significantly better than students and advanced students, while in Functioning and Evaluation both librarians and advanced students perform better than students in the first years. Librarians' total weighted score (WNS) (avg. = 68.1, sd = 11.4) is significantly better than both students (avg. = 58.2, sd = 13.9) and advanced students (avg. = 61.9 sd = 14.7).

The three groups differ also in their distinct patterns of AI literacy levels (Fig. 4). Librarians show the highest proportion of high literacy (with 49 %), followed by 43 % at the medium level and only 8 % at the low level. Among advanced students, 58 % demonstrate a medium level of literacy, 25 % reach a high level, and 17 % fall into the low category. In contrast, undergraduate students present a more polarized

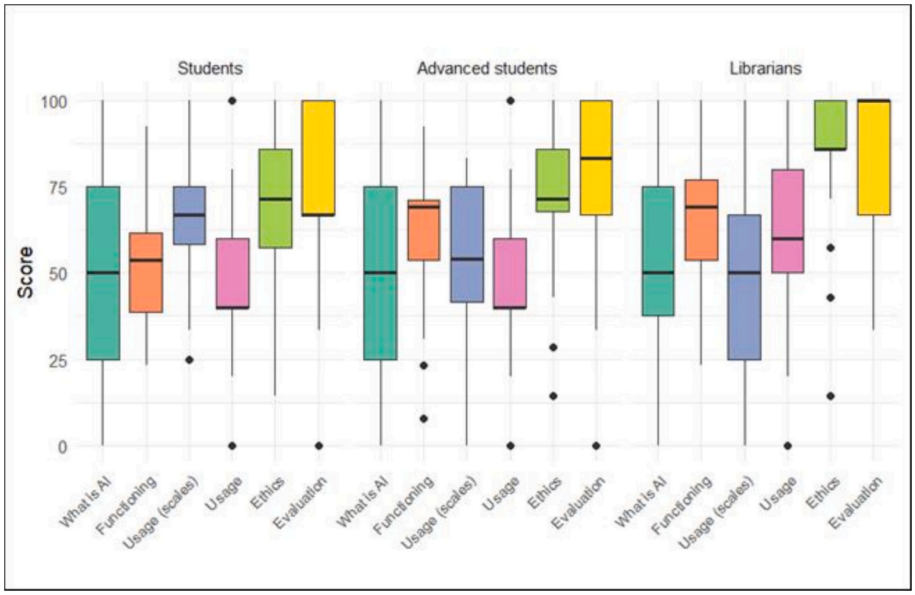

Fig. 3. Scores obtained by two different groups of students and librarians.

Table 5
Kruskal-Wallis test results for comparison of students, advanced students and librarians.

Dimensions	Kruskal-Wallis Test Results
What is AI	Non significant
Functioning	Kruskal-Wallis chi-squared = 14.674, df = 2, *p*-value = 0.0006509
Usage (scales)	Kruskal-Wallis chi-squared = 25.89, df = 2, p-value = 2.388e-06
Usage	Kruskal-Wallis chi-squared = 15.615, df = 2, p-value = 0.0004067
Ethics	Kruskal-Wallis chi-squared = 35.191, df = 2, p-value = 2.283e-08
Evaluation	Kruskal-Wallis chi-squared = 6.4103, df = 2, p-value = 0.04055
WNS	Kruskal-Wallis chi-squared = 18.254, df = 2, p-value = 0.0001087

distribution, with 52 % at the medium level, 27 % at the low level, and 21 % achieving high AI literacy. These results suggest that librarians tend to exhibit higher AI literacy than students, particularly in the high-level category.

Gender differences

By gender, there are small differences in average scores, generally favoring female participants (Table 6). However, the Wilcoxon rank-sum test did not yield statistically significant results, so the null hypothesis could not be rejected. Participants who did not indicate their gender were excluded from the test due to the small size of this subgroup (n = 7).

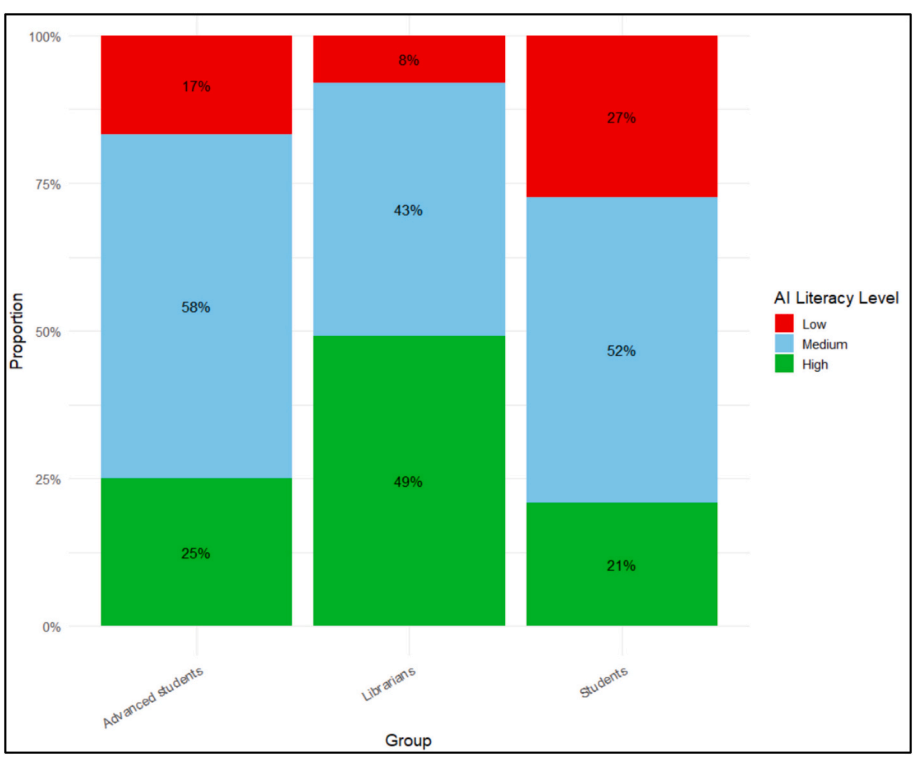

Fig. 4. AI literacy score distribution by group and proficiency level (High/Medium/Low).

Table 6
Average scores by gender across different dimensions.

	Females		Males		Prefer not to tell	
Dimensions	Average	SD	Average	SD	Average	SD
What is AI	51.4	27.0	54.9	26.0	57.1	18.9
Functioning	61.8	18.6	60.2	18.1	56.0	18.7
Usage (scales)	54.0	24.4	53.3	24.3	65.5	11.2
Usage	57.9	23.7	53.3	23.6	51.4	19.5
Ethics	76.6	19.1	70.6	24.0	81.6	19.7
Evaluation	78.0	25.8	70.6	28.0	76.2	25.2
Total Weighted Score	63.8	13.6	61.0	14.5	63.3	13.1

Female scores exhibit greater individual variability than male scores across most dimensions, with the exception of Ethics and Evaluation (Fig. 5). The wider spread observed in technical dimensions and in Usage reflects greater internal diversity in AI literacy levels among women.

Previous training

Regardless of participants' role, most have not received previous training, with the highest proportion among librarians (Table 7). Participants who received some form of training - whether in the classroom, through formal courses, or via self-study - achieved significantly higher total scores (W = 2267.5, p = 0.014), as well as higher scores in the Functioning (W = 2270, $p = 0.014$) and Usage (scales) dimensions (W = 2395, $p = 0.043$). On average, their scores exceeded those of participants with no training by 5.5, 7.6, and 9.2 points, respectively, suggesting that training has a greater effect on self-efficacy than on areas of AI literacy that were measured based on performance.

Themes extracted from focus groups

Results obtained from the questionnaire were further validated by students' feedback. Four themes were identified from the analysis of the focus groups: (1) human–machine interaction; (2) unproblematic aspects of AI usage; (3) problematic aspects of AI usage and negative reactions; and (4) generational inequalities.

The four themes highlight students'- and indirectly, through their reflections, educators'- lack of preparedness for AI, and a general underestimation of its complexity due to the human-like and accessible interfaces that it offers. Students' experiences indicate that, despite its apparent ease of use, engaging with AI in academic settings in a thoughtful and constructive manner demands careful evaluation, time, and sustained effort. Students also feel that developing AI literacy requires ongoing practice, real-world application, and positive, effective experiences with the technology.

Focus groups validate questionnaire results, as *functioning* emerges as

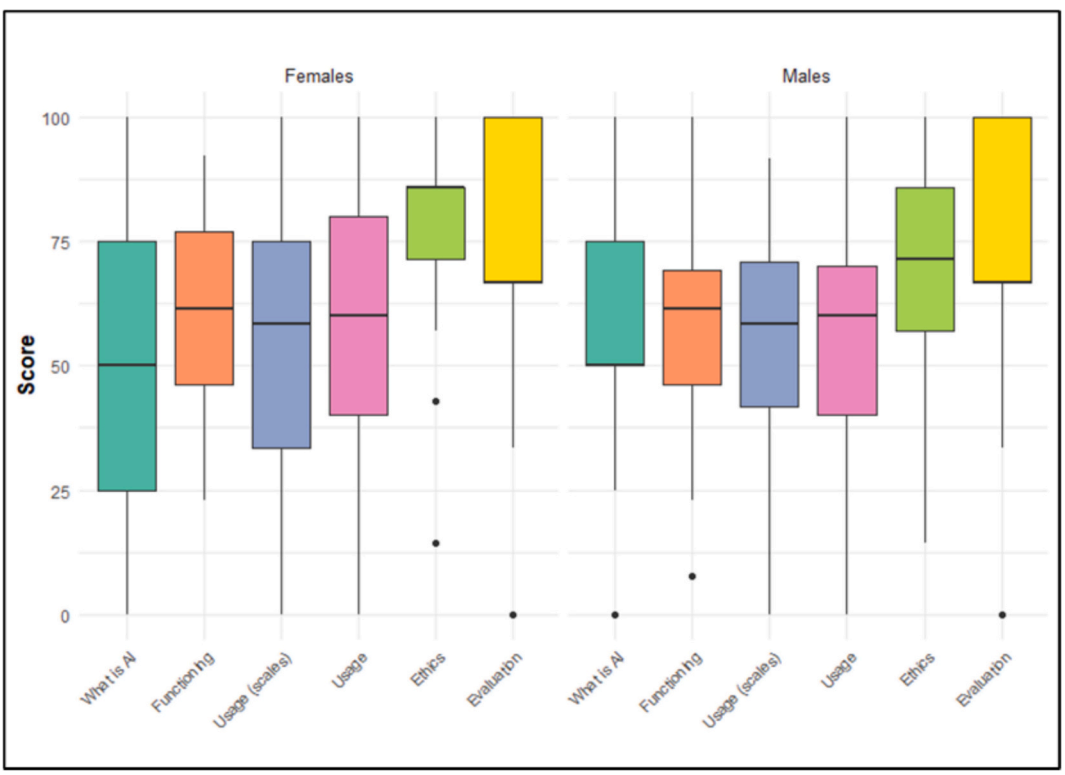

Fig. 5. Scores obtained by female and male participants.

Table 7
Previous AI training across different roles.

Roles	*Has not received tranining*	*%*	*Trained through self-study*	*%*	*Has attended formal training*	*%*	*Has received class-room training*	*%*	*Total*
Students	*40*	*64,5 %*	*2*	*3,2 %*	*2*	*3,2 %*	*18*	*29,0 %*	*62*
Advanced Students	*23*	*63,9 %*	*5*	*13,9 %*	*0*	*0,0 %*	*8*	*22,2 %*	*36*
Librarians	*43*	*68,3 %*	*17*	*27,0 %*	*3*	*4,8 %*		*0,0 %*	*63*
Others	*2*	*100,0 %*		*0,0 %*		*0,0 %*		*0,0 %*	*2*
	108	*66,3 %*	*24*	*14,7 %*	*5*	*3,1 %*	*26*	*16,0 %*	*163*

the AILIS 1.0 dimension most in need of further training, while *self-efficacy* proves to be a poor proxy for actual AI literacy, as many participants express surprise at their lower-than-expected performance. In all focus groups, participants point to AI functioning as the area where they feel least prepared, as P. explains: "*So, when there were more technical questions, it was like… I could understand some of the answers, but I wasn't really sure.*" Nearly all express surprise at the low scores they got, having overestimated their actual knowledge and competencies. This overestimation may be related to the friendly and intuitive nature of chatbot interfaces, which give a false sense of mastery. So it emerges from the summary of Focus group n° 3, as *"…they found it [the questionnaire] 'long and complicated' … they expected it to be a quick questionnaire (one that wouldn't require much thought), and the fact that it forced them to 'reflect' was something 'new and challenging' for them. Some sighed."*

Human–machine communication

Students repeatedly emphasize different aspects of communication with AI, and the theme of human–machine interaction is central in all their accounts. They are aware that communicating with AI requires the ability to construct meaningful prompts - a type of communication they are not yet accustomed to - as J. notes when asked for possible improvements of the questionnaire: *"How to construct prompts properly… like, how to write sentences so that the response is much clearer and not so vague. That's something new…"* They also acknowledge the importance of evaluating AI outputs and filtering out irrelevant, noisy, or misleading information, recognizing that effective communication with AI is not spontaneous but demands both time and careful consideration. In N.'s words: *"…at first, you read it [an AI response] and it makes sense, but then when you get into the details, it has nothing to do with what you actually asked."*

Even though AI is often humanized and perceived as almost human - *"It is almost like a psychologist,"* L. remarks - students are aware that communicating with machines differs from human interactions in several ways. In particular, they find that it is difficult to assess the trustworthiness of the information provided, which increases uncertainty and often leads to breakdowns in communication when AI fails to provide the expected answers. As A. remarks, its unstable responses contribute further to frustrate many attempts of communication: *"… I personally also feel a bit of distrust toward ChatGPT, because we've also been warned: 'don't fully trust it because it makes up references', 'it makes up all sorts of things'… So I ask it things that I know are true, that I know how they really are… Then I ask: 'Why did you lie to me?' And it replies, 'Oops! My apologies!'"*

Students are aware that AI tools are trained on different knowledge bases and can thus provide different responses, but what confuses them most are the inconsistencies in AI's responses across different interactions, pointing to AI unstable and opaque knowledge bases. For instance, when using AI chatbots to answer class quizzes, students sometimes receive different answers from the same tool on different computers, which puzzles and confuses them.

Unproblematic aspect of AI usage

Curiosity to explore and better understand AI appears in the accounts of some students, and reflects a positive, optimistic view of AI. As Sa. comments: *"It'll be fine if we use it correctly."* This attitude typically emerges in connection with successful experiences using AI - particularly for tasks related to text processing and code generation - both in academic and professional contexts, as many of the students were completing internships. Ev. tells her professional story: *"At work, we're indexing some journals that we have digitized as PDFs… So what I do is take a screenshot of the PDF page and feed it into Gemini. I tell it 'summarize,' and it gives me a summary of the article. But Gemini isn't very accurate… it usually gives you a super long summary. So, since I also have to extract keywords, I take that Gemini summary and put it into ChatGPT, and it gives me a shorter summary and the keywords. A task that used to take me two hours it's now taking me 45 minutes or less…"*.

Such successful experiences often lead to a more optimistic outlook on future professional scenarios, as expressed by E.: *"I think that artificial intelligence in its own time, will settle into the place where it belongs. Because back in the day, it was the same with Google… but over time, things find their place, and people will naturally put it where it needs to be."* Notwithstanding, AI also provokes a wide range of more negative reactions, which leads into the third theme identified in the focus group analysis.

Problematic aspects of AI and negative reactions

Being increasingly present in the academic and professional lives of students, AI brings with it a series of problematic aspects and related emotional responses. Uncertainty, alongside frustration, confusion, and guilt, characterizes this theme, which focuses on the challenges AI presents. Coexisting with humanized or human-like agents such as AI calls into question participants' personal and professional identities, especially given the constant need to adapt to rapidly evolving technologies and the emergence of new commercial tools.

L. worries about the possibility of her voice being cloned in a phone call, which threatens her personal identity, whereas T. wonders what makes her different from other professionals, as anyone can now perform many specialized tasks for which she is being trained. AI can assist with hard tasks such as coding or interpreting technical texts. However, this also means that anyone can now perform such specialized tasks: *"Today, for example, at work, during my internship, right? They gave me a document that came from a notary, and they write in such a technical way that I didn't know which title to give it… So, I turned to ChatGPT, because I've gotten so used to it… and that's a shame, because someone who has studied this degree is now competing with someone who hasn't studied it… well, they should choose this person, not someone who uses ChatGPT… Otherwise, anyone could do my job…"* (T.)

Not only are students' identities as individuals and future professionals challenged and threatened, but they are also aware that they are now competing from an unequal position with human-like information systems capable of performing many of their intended job tasks more efficiently - and without questioning the nature of what they are asked to do. As a result, students anticipate a more competitive and demanding labor market, as S., A., and N. remark:

"S.: In the end, it's going to be a more competitive job market…

A.: And not just between people… but also with machines.

N.: Competing with something that's not a person and doesn't think, but it just does what you tell it to…"

This unequal position also makes them aware of the growing dependence among young people, including themselves, on various AI tools used in all areas of life. Many begin by using AI for academic tasks and soon find themselves relying on it for everyday decisions, such as planning meals or gym routines. *"Or for getting organized, when it comes to studying. You can say, for example, 'I have five days to study for an exam, structure everything I need to study'… before, we used to make our own study calendars,"* M. adds. The potential risk is to stop thinking independently of the machine, as A. remarks: *"If you delegate such basic things like a menu and say, 'Make me the full macaroni recipe,'… in the end, you don't think for yourself…"*

This growing dependence, the loss of independent thinking, and the perception of professional threats lead students to feel at real risk of intellectual and career-related impoverishment. Ch. puts it this way: *"We can't let artificial intelligence take away our opportunities for mental exercise."*

The need to adapt to a fast-changing AI landscape, and the effort it demands, leads to diverse emotional reactions. Some students reject AI altogether or deny its real impact, adopting an attitude of what could be termed *"AI negationism"*, with judgments such as "*I don't think I'll ever do it better*" (S.), "*it makes it up*" (C.), "*it mixes things that aren't related*" (C.), *"it doesn't do it well"* (An.), or *"it messes up a lot."* (An.) Many are confident that the impact of AI will take time to materialize. *"It worries me in the long term, maybe in a few years"* says An., while S. comments: *"Right now, none of this is required for the exams, so I'm not worried."*

Dependence also amplifies feelings of uncertainty, frustration, and confusion, with guilt surfacing when students use AI to simplify academic work, even being aware that this may reduce their learning opportunities or be unethical. Sometimes guilt appears expressing a broader sense of responsibility, especially of educators and adults, toward younger generations. M. gets upset at N.'s story:

"N.: My brother is in 10th grade… the other day he had to do a bibliography, and the teacher told him to do it with ChatGPT.

M.: The teacher said that?!"

Generational inequalities

Older generations' responsibilities are only timidly brought to the fore, as participants primarily spoke to their own teachers. More often, examples from outside their immediate educational environment are cited to highlight older generations' responsibilities, including cases of misconduct in scientific communication or anecdotes such as the one previously mentioned by N. While participants recognize that AI impacts all generations, they understand that the burden is unevenly distributed, with the youngest bearing the greatest cost, and the oldest not doing enough. The intellectual and career-related impoverishment they perceive represents a clear generational inequality, although they also foresee similar challenges for even younger people, such as family members or acquaintances in secondary or primary education. Children can easily complete many learning tasks using AI, unless there is close supervision or active involvement from educators: *"Summaries of books that maybe you haven't even read.... now you can do that with ChatGPT. The issue is knowing how it will be approached - not banning it or ignoring it or teaching it later, but how. How are you going to teach it to kids?"* (S.). A. worries that children may be becoming *"less intelligent"* or that *"they just don't try as hard."* On the other hand, the ability of AI technologies to permeate all areas of life makes J. wonder whether the youngest generation will still be able to communicate directly with one another: *"And combined with social media, kids… well, they might not even know how to communicate directly anymore."*

Discussion

Limitations

Many limitations of this study relate to the instrument used to measure AI literacy, namely AILIS 1.0. Firstly, the process of transforming competencies into questionnaire items may have introduced bias or subjectivity, which should be acknowledged as a limitation of the instrument. However, this approach also enabled measurement of AI literacy through a performance-based scoring system, differing from most other AI tools that rely primarily on self-assessment, with significant implications for understanding AI literacy, as discussed later. Secondly, the performance-based nature of the instrument makes it more susceptible to becoming outdated particularly given the rapid evolution of AI technologies. In fact, the administration of the questionnaire was suspended in March 2025 due to some items becoming obsolete.

A second set of limitations concerns the sample, which, while small and diverse, including both LIS students and professionals, is nonetheless representative of the target population – especially UCM LIS students. This may limit the generalizability of the findings to broader LIS contexts.

The comparison of the test and retest results provided moderate to poor reliability coefficients, especially compared to previous research into AI literacy. However, in most AI instruments, reliability is often measured as internal reliability (Cronbach's alpha), which does not imply a procedure of test and retest, thus omitting the stability of the results over different administrations (De Vet et al., 2011; Leppink & Pérez-Fuster, 2017). On the other hand, the need for the reliability and precision of tools like AILIS 1.0 grows as the consequences of decisions based on test results become more significant (American Educational Research Association (AERA) et al., 2014: p.33). In this study, questionnaire results are intended to identify areas in need of further training, with the additional support of feedback provided by students in three focus-groups, which makes moderate to poor reliability coefficients acceptable.

Finally, participants' experiences and perceptions were collected exclusively from students, and not from librarians. The librarians' perspective is expected to be incorporated in future stages of the research.

Which areas should AI training in LIS pay attention to?

Pinski and Benlian (2024) highlighted the need to identify AI-specific competencies within AI literacy, given its significant overlap with other forms of literacy. Both the analysis of questionnaire results and the focus groups point to Functioning, Ethics, and Evaluation as core areas of AI literacy in LIS. Functioning has emerged from previous research as a dimension in which higher education students perceive to be less prepared, except students from technical areas (Hornberger et al., 2023; Hossain et al., 2025; Lee et al., 2024; Toker Gokce et al., 2024). In this research, Functioning was given relevance in the design of a questionnaire specifically addressed at a LIS audience, with particular emphasis on data-related competencies of Functioning, which differentiate it also from other frameworks intended to measure AI literacy in LIS (LaFlamme, 2025; Lo, 2024; Shiri, 2024). In this sense, AILIS 1.0 also reflects the shifting emphasis from text to data in library services (Cox, 2023). Additionally, previous frameworks for AI literacy in libraries have focused on softer skills (Ethics and Evaluation) more in line with information literacy frameworks and library traditions. This research, on the other hand, points to the need to cultivate harder competencies, especially regarding the technical functioning of such technologies.

Results revealed that better scores in Functioning were associated with better scores in all other dimensions, especially Ethics and Evaluation, except the Usage dimension measured as perceived efficacy (Usage - scales). Traditionally, self-efficacy in technology usage has been considered a good predictor of successful technology use. In the case of AI technologies, however, it seems that actual knowledge of AI functioning can be a good predictor of successful AI usage, at least according to the results of this research and some previous studies (Tzirides et al., 2024). AI explainability and knowledge of its functioning is an essential skill for human-AI co-creation activities (Wen et al., 2024), and this research supports understanding of how AI works and functions as an essential asset for AI literacy. Additionally, the findings suggest that participants who had received AI training—about one-third of both students and librarians—achieved better outcomes than those without training. The improvement was most notable in self-assessed usage (scales), reflecting higher self-efficacy, and was also evident in the Functioning dimension, indicating that training can positively impact a core area of AI literacy and reverberate across related competencies. Finally, the focus groups confirmed a tendency to overestimate actual AI literacy, particularly among less advanced students, who reported the highest self-assessed competencies but scored lowest on performance-based measures. This may stem from the human-like and user-friendly interface of AI tools, which can be misleading and give a false sense of mastery due to their apparent ease of use. This tendency to overestimate actual AI knowledge and competencies raises questions about the reliability of self-assessment in measuring AI literacy.

The importance of a strong technical understanding of AI for AI literacy is only partially addressed in existing LIS literature, despite evidence of librarians' limited proficiency in more technical areas such as AI programming (Paladhi & Maruthaveeran, 2024). In Bongiovanni et al. (2024), the topics that academic librarians currently cover or plan to cover in AI training primarily focus on issues such as AI limitations, ethics, plagiarism, critical evaluation of AI tools, bias, misinformation, hallucinations, and false citations, with more technical aspects of AI in the background. In the study by Hossain et al. (2025), students placed greater emphasis on the ethical implications of AI use, despite having

limited understanding of how the underlying machine learning models (MLMs) function. In contrast, Lo's survey (2024) highlights that a solid foundational understanding of AI is essential for setting realistic expectations and making informed decisions regarding its implementation in library settings. However, while greater attention should be given to the technical functioning aspects of AI literacy, it is equally important to keep nurturing softer skills. Meaningfully, reliability scores were especially poor in Ethics and Usage, that can be considered softer dimensions, likely pointing to the need of clearer guidelines and targeted training on AI integration in higher education. Poor reliability in these dimensions was supported by reported feelings of uncertainty, frustration, confusion, and guilt in focus groups when participants discussed their use of AI for educational purposes, with implications for their professional identities. Previous research has already underscored students' need for clear guidelines on AI ethics and appropriate usage (Chen et al., 2024; Daly et al., 2024). The absence of explicit guidance can also undermine students' professional identity, as Vázquez-Parra et al. (2024) found that students from diverse disciplinary backgrounds recognized the professional relevance of AI tools, yet expressed ambivalence about their formal inclusion. However, it also points to a possible educators' delay in integrating AI into syllabi and daily teaching. A recent OpenAI report on AI usage in higher education (2025) highlights the uneven involvement of universities in training students to use AI. Although the report focuses primarily on institutions in the United States, its findings are also relevant to the Spanish context, where this research was conducted, and potentially to others. Not only did students feel especially uncertain about actual AI applications and usage, but they were also worried about the ethical implications of this scenario. The skepticism toward AI outputs previously observed among higher education students (Fleischmann, 2024) and LIS professionals (Cox, 2023) emerged in this study intertwined with a complex mix of negative emotions, which in some cases led to a form of "AI negationism." In this sense, focus groups revealed that students feel that, as younger generations, they are in an unequal position in regard to older generations. Professionally, they face a more competitive labor market in which many others can perform tasks traditionally considered specific of their area of specialization. Educationally, they often lack motivation to engage in tasks that AI can now perform more efficiently, thereby missing valuable formative opportunities. Mühlhoff (2023) discusses informational power asymmetries between platform companies and users, whose data are increasingly analyzed for profiling and commercial purposes, providing evidence of students' pervasive feeling of being unequally positioned in this new technological landscape. Special concerns were expressed about dependence on AI, loss of independent thinking, and a general "impoverishment" both in intellectual abilities and career-related matters. Huang (2023) warns about the consequences of the close relationship between people and machines with human actions being constantly tracked in digital systems. However, from the perspective of students, it is not only personal data they are deprived of, but also possibilities to grow intellectually, and opting to satisfactory professional positions in which their specialized training is valued and recognized. The ethical concerns that emerged from this study are partly different from those outlined by LIS previous research. Specifically in relation to students, for instance, Hossain et al. (2025) stressed plagiarism and academic integrity over more global generational and societal issues. Among LIS professionals surveyed by Lo (2024), the primary ethical issues were related to privacy concerns and misuse. Conversely, Cox (2022) provides a more comprehensive view of ethical issues related to AI, including those concerning sustainability and professional implications.

Differences among participants

Differences among groups of participants taking the questionnaire reveal the effect of training on AI literacy, as previously commented on, additionally demystifying the assumption of a female disadvantage in technology use. Using a performance-based tool to measure AI literacy, no significant differences were found between male and female participants' completion scores, despite expectations based on previous research on AI literacy and attitudes toward AI (Grassini, 2024; Sindermann et al., 2021). However, greater variation in individual performances was observed within the female group, indicating higher diversity in AI literacy levels among women. Within the LIS field, Hossain et al.'s (2025) survey found that male students reported higher levels of AI literacy. However, the authors cautioned that these findings may be influenced by self-assessment bias. In all cases, it appears that a mixed scoring system would be preferable when measuring AI literacy in order to grasp a more realistic view of actual AI knowledge and competencies across different demographic groups. Also, librarians turned out to be better prepared in all dimensions than first-years students, while they showed a more realistic view of their actual Usage competencies - measured as Usage (scales). With half of the librarians achieving high scores on the test, their level of literacy was shown to be substantially higher than that of the students. On the other hand, they obtained significantly better scores than advanced students only in the Usage and Ethics dimensions. This finding should encourage library professionals to take a more active role in delivering AI training to higher education students, especially in their initial years, given that such training has, to date, been uneven and limited, as highlighted in the introduction (Bongiovanni et al., 2024). A greater involvement of librarians in AI training within higher education could also position them as role models for future generations of professionals in the field.

Conclusions

In conclusion, this report's analysis of questionnaire data and focus group discussions provides insights for improving nearly all dimensions of AILIS 1.0, and more broadly, advancing AI literacy in higher education and in academic libraries. In general, AI literacy in LIS emerges as a construct that should not be conceived simply as an extension of other forms of literacy, to avoid underestimating its effect on LIS higher education and professions. Also it should not rely solely on ethical awareness and information literacy traditions, but should be complemented by stronger technical and data-related competencies. More specifically, regarding the "What is AI" dimension, it is crucial that students and users recognize that, despite their human-like interfaces and apparent ease of use, AI systems are complex technologies driven by human-generated data - including their own. Maintaining this critical awareness is essential to approach AI not merely as passive "consumers," but as active "creators" and "innovators," echoing the terminology used by Chang and Kuo (2025). A clearer understanding of AI's true nature is also necessary to cultivate a more comprehensive and conscious understanding of users' roles in interacting with it, moving from mere usage to collaborative engagement and "co-creation" (Wen et al., 2024). This can also prevent overestimation of one's knowledge and competencies, while encouraging realistic expectations for its responsible and constructive usage. Evaluating AI-generated content, in this sense, demands time and effort, and users should avoid treating AI as a quick-and-dirty shortcut for addressing academic or professional challenges, even if it can dramatically improve many routine tasks. With regard to the "Functioning" dimension, AI literacy needs to emphasize the rapid pace of technological change and the ongoing need for users to regularly update their knowledge and skills. Finally, in the area of Ethics, it is important to explicitly address generational inequalities, the responsibilities of educators and older generations, as well the risk of intellectual and professional impoverishment. This will more likely ensure that students, but also educators and librarians grasp what is truly at stake, because, as Bennett and Abusalem (2024) contend, the current university model may be no longer fit-for-purpose.

For all these reasons, AILIS 1.0 proves to be a promising tool not only for assessing current levels of AI literacy, but also for identifying blind spots in curricula and for guiding the design of targeted educational

interventions. The study also calls for institutional strategies that formally recognize and support AI literacy training. Librarians, as demonstrated by their higher performance and more balanced self-perception, can be mentors, bridging the gap between academic training and professional practice.

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CRediT authorship contribution statement

Michela Montesi: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Belén Álvarez Bornstein:** Validation, Supervision, Resources, Investigation, Funding acquisition, Conceptualization. **Núria Bautista Puig:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Manuel Blázquez Ochando:** Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Alicia Sánchez Díez:** Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Michela Montesi reports a relationship with Journal of Academic Librarianship that includes: board membership. Given Michela Montesi's role as member of the editorial board of the Journal of Academic Librarianship, she had no involvement in the peer review of this article and had no access to information regarding its peer review. Full responsibility for the editorial process for this article was delegated to another journal editor. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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